ECE C147/C247, Winter 2024 **Midterm Examination**
Neural Networks & Deep Learning Prof. J.C. Kao
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UCLA True Bruin academic integrity principles apply.
Open: Four pages of cheat sheet allowed, calculator.
Closed: Book, computer, internet.
2:00pm-3:50pm PST.
Wednesday, 21 Feb 2024.

State your assumptions and reasoning.
No credit without reasoning.
Show all work on these pages.

Name: _____

Signature: _____

ID#: _____

Problem 1	_____ / 20
Problem 2	_____ / 35
Problem 3	_____ / 20
Problem 4	_____ / 25
BONUS	_____ / 7 bonus points
Total	_____ / 107 points

1. **Multiple choice and short answer.** (20 points)

   For multiple choice questions, select all true statements that correctly answer the question (there may be multiple). You do not have to justify your answers for multiple choice questions.

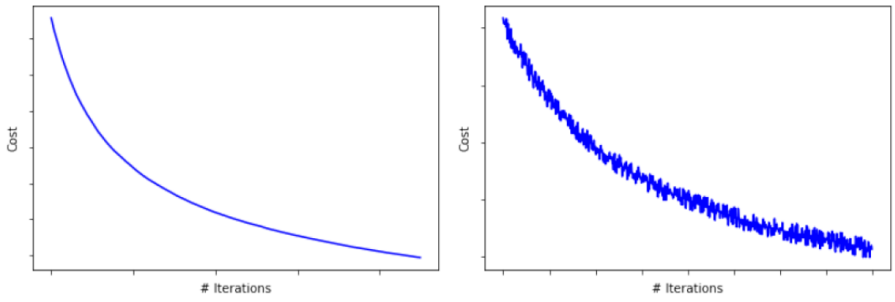
   (a) (3 points) You are tasked with training a 2 layer Fully Connected Neural Network (FCNN) with batch normalization and dropout. Which of the following are hyperparameters?

   i. Loss function
   ii. Dropout probability p
   iii. Batchnorm rescaling factors γ and β
   iv. Batch size

   (b) (3 points) Which of the following are valid techniques to perform regularization?

   i. Early Stopping.
   ii. Norm Penalties.
   iii. Model Ensembling.
   iv. Dropout.

   (c) (3 points) Consider the training loss curves, shown in Fig. 1, for two different optimization routines. In one routine we estimate the gradient using a Batch (entire training set) and in the other we estimate the gradient using a Minibatch. Based on these graphs, which of the following are true?



(a) Graph A (b) Graph B

Figure 1: Loss trajectory of two different optimization routines

   i. Graph A represents Minibatch Gradient Descent since the gradient estimates using batch and mini-batch are equivalent.
   ii. Graph A represents Batch Gradient Descent because it estimates the gradient accurately, resulting in a decrease in loss at every iteration.
   iii. Graph B represents Minibatch Gradient Descent since it is a noisy estimate of the gradient, resulting in oscillations in the loss trajectory.

(d) (3 points) Select the true statements below regarding k-fold cross validation.

i. We evaluate the cross validation performance as the mean of the performance on the validation set of the k different folds.

ii. We evaluate the cross-validation performance as the best one over k different folds.

iii. In a k-fold cross validation approach, k different model instances are fitted.

iv. In a k-fold cross validation approach, the same model instance is applied to evaluate the validation performance of all folds.

(e) (5 points) Your friend encounters problems when using a Fully Connected Neural Network (FCNN) for a deep learning task and asks for your help.

i. (2 points) Your friend is confused about whether to choose ReLU or PReLU as the activation function. Below are the equations for these two activation functions, where α is a learnable parameter.

$$\text{ReLU}(x) = \max(0, x), \qquad \text{PReLU}(x) = \begin{cases} x, & \text{if } x > 0 \\ \alpha x, & \text{otherwise} \end{cases}$$

Please explain one advantage of PReLU over ReLU in no more than three sentences.

ii. (3 points) In the experiment, the FCNN model achieves a low training error and a high test error. Your friend thinks this is due to underfitting and plans to add more layers to increase the model capacity. Do you agree with his evaluation? Justify your answer and, if you disagree with your friend, provide two suggestions to improve model performance.

(f) (3 points) Suppose you are deploying Convolutional Neural Networks (CNNs) for image classification. Describe two advantages of a CNN over an FCNN in no more than 3 sentences.

2. **Hybrid norm constraints in Similarity Neural Network (SNN)** (35 points)

A similarity neural network (SNN) consists of twin networks which accept distinct inputs but share the same weights. The outputs of the twin networks are later processed by more hidden layers. Let's assume we have a SNN with the following architecture:

$$
\begin{aligned}
h_p &= W_1 x_p^{(i)} + b_1 \\
z_1 &= \text{softplus}(h_p) \\
h_q &= W_1 x_q^{(i)} + b_1 \\
z_2 &= \text{softplus}(h_q) \\
z &= z_1 - z_2 \\
z_3 &= W_2 z + b_2 \\
\hat{y}^{(i)} &= \sigma(z_3) \\
L^{(i)} &= L_{CE}(y^{(i)}, \hat{y}^{(i)}) \\
L &= -\frac{1}{m} \sum_{i=1}^{m} L^{(i)}
\end{aligned}
$$

In the above architecture, $(x_p^{(i)}, x_q^{(i)})$ represents the pair of ith input example and are each of shape $\mathbb{R}^{D_x \times 1}$. $y^{(i)}$ represent the label of the ith input example and is a scalar, $y^{(i)} \in \{0, 1\}$. We also assume z_1 and z_2 have shape of $\mathbb{R}^{D_z \times 1}$. $\sigma(\cdot)$ denotes the sigmoid function. L_{CE} denotes the binary cross entropy loss. Softplus is a smooth approximation to the ReLU function and can be used to constrain the output to always be positive. It is an element-wise function defined as follows:

$$\text{softplus}(x) = \frac{1}{\beta} \log(1 + e^{\beta x})$$

For your convenience, we have drawn out the computational graph for you in Figure 2.

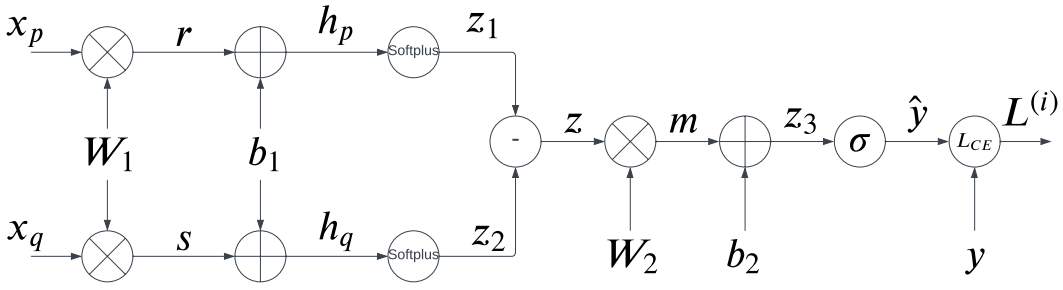


Figure 2: Computational graph for SNN

(a) (3 points) Compute $\nabla_{\hat{y}^{(i)}} L^{(i)}$ and denote it as $\delta_{\hat{y}^{(i)}}$. For all the following parts, you can refer to this computed gradient as $\delta_{\hat{y}^{(i)}}$.

Hint: Recall the binary cross entropy loss is:

$$L^{(i)} = y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})$$

(b) (3 points) Compute $\nabla_{z_3} L^{(i)}$ and denote it as δ_{z_3}. For all the following parts, you can refer to this computed gradient as δ_{z_3}.

(c) (3 points) Compute $\nabla_{W_2} L^{(i)}$ and denote it as δ_{W_2}. For all the following parts, you can refer to this computed gradient as δ_{W_2}.

(d) (3 points) Compute $\nabla_z L^{(i)}$ and denote it as δ_z. For all the following parts, you can refer to this computed gradient as δ_z.

(e) (4 points) Compute $\nabla_{h_q} L^{(i)}$ and denote it as δ_{h_q}. For all the following parts, you can refer to this computed gradient as δ_{h_q}.

(f) (4 points) Compute $\nabla_{h_p} L^{(i)}$ and denote it as δ_{h_p}. For all the following parts, you can refer to this computed gradient as δ_{h_p}.

(g) (2 points) Compute $\nabla_{b_1} L^{(i)}$.

(h) (5 points) Compute $\nabla_{W_1} L^{(i)}$.

(i) (8 points) Based on some prior knowledge, you want to impose some norm constraints on W_1 and W_2. After exploring the various norm constraints, you have decided to modify the loss function in the following way:

$$\tilde{L} = \left(-\frac{1}{m} \sum_{i=1}^{m} L^{(i)} \right) + LSE(W_2) + \|W_1\|_\infty,$$

where Log-Sum-Exponent (LSE) of a vector $x \in \mathbb{R}^n$ is defined as follows:

$$LSE(x) = \ln \left(\sum_{i=1}^{n} e^{x_i} \right)$$

Compute $\nabla_{W_1} \tilde{L}$ and $\nabla_{W_2} \tilde{L}$.

3. **Regularization techniques** (20 points)

(a) (7 points) Derive the gradient descent step for linear regression where $\mathcal{L}(\theta) = \frac{1}{2}(\mathbf{y} - \mathbf{X}\theta)^T(\mathbf{y} - \mathbf{X}\theta)$ where $\mathbf{y} \in \mathbb{R}^N$, $\mathbf{X} \in \mathbb{R}^{N\times 2}$, and $\theta \in \mathbb{R}^2$.

Hint: $\frac{\partial \mathbf{z}^T\theta}{\partial \theta} = \mathbf{z}$ and $\frac{\partial \theta^T\mathbf{A}\theta}{\partial \theta} = (\mathbf{A} + \mathbf{A^T})\theta$.

(b) (5 points) Derive the gradient descent step for ridge regression where $\tilde{\mathcal{L}}(\theta) = \mathcal{L}(\theta) + \frac{\alpha}{2}||\theta||_2^2$.

Hint: You can leave your final answer in terms of $\nabla_\theta \mathcal{L}(\theta)$.

(c) (4 points) Explain one regularizing effect of ridge regression on a model. Your answer should not exceed three sentences.

(d) (4 points) You implemented inverted dropout in Homework 4. What is the reasoning behind dividing the activations by p, where p is the probability of keeping a neuron, in the forward pass of training? Your answer should not exceed three sentences.

4. **Optimization techniques** (25 points).

   (a) (7 points) Consider the loss function with a local minima as shown in Figure 3. We assume we started at the initial position labeled 'Start' and arrived at a local minima after 5 iterations of vanilla gradient descent as shown in Figure 3. We have accumulated some momentum along the way.

   After arriving at the local minima, we are considering the next gradient descent step, which we will take with the accumulated momentum. We have two options: Option 1: GD + momentum, and Option 2: GD + Nesterov momentum. Between Option 1 and 2, which optimizer will result in a weight update with larger magnitude? Explain your answer in no more than 4 sentences. If your answer depends on the accumulated momentum, state why that is in your answer.

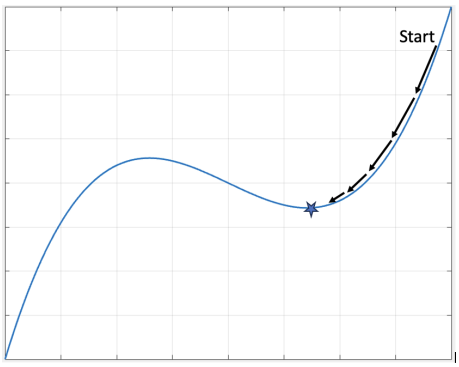


Figure 3: Loss function $L(x)$

(b) (10 points) Consider the contour plot shown in Figure 4 , where the level sets of the loss surface is plotted with respect to the 2 weights X and Y, where $X, Y \in R$ (scalars). From the contour plot, we can tell that the shown region contains a "valley". Assume we are given a hypothetical scenario: we start from point 1 and use vanilla gradient descent for many iterations to get to point 2. During this process, we accumulated momentum (first moment) and second moments. Now carefully observe the plot, and answer the following questions for the next few iterations of weight updates.

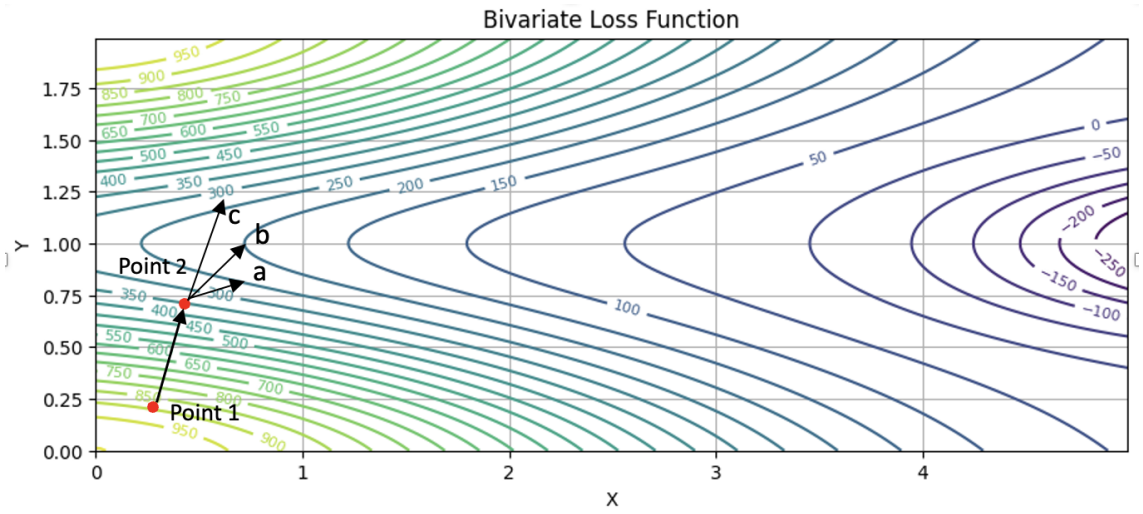


Figure 4: Contour plot of a Loss function $L(X, Y)$

i. (7 points) The update steps a, b and c in the contour plot, correspond to three different algorithms based on the same history trajectory, are plotted. Please match the steps with the following three optimization algorithms: GD + Momentum, Adagrad and Adam. Justify your reason in no more than 7 sentences.

ii. (3 points) Which of the above three algorithms will face issue in escaping the valley in the next few iterations ? Explain the reason for this issue.

(c) (8 points) In the Gradient descent + Nesterov momentum scheme, find a general expression for θ_t (weight parameter) in terms of θ_0 (initial weights), the gradients $g_1, g_2, ..., g_t$, α (momentum hyperparameter) and ϵ(learning rate), considering an initial value of momentum $v_0 = 0$. Consider the following equations for the weight update in Nesterov momentum:

$$v_t = \alpha v_{t-1} - \epsilon g_t$$
$$\theta_t = \theta_{t-1} + v_t + \alpha(v_t - v_{t-1})$$

5. **Bonus: Sparsity using ℓ_1 regularization** (7 points)
Please note that no partial credit is awarded for the bonus problem

In lectures and discussion, we have remarked that adding an ℓ_1 regularized term promotes sparsity in the weight vectors. In this problem, we will show the mathematical justification for the remark. Let's consider the following regualrized cost function:

$$L_{reg}(w) = L(w) + \alpha\|w\|_1.$$

Applying Taylor series expansion to $L(w)$ near the optimizer w^* and discarding higher-order terms we get

$$L(w) \approx L(w^*) + \frac{1}{2}(w - w^*)^T H(w - w^*),$$

where

$$w^* = \arg\min_w L(w)$$

and $H = diag(h_1, h_2, \cdots, h_n)$ is a diagonal matrix. You can assume $h_i > 0, \forall i$.

(a) (2 points) Use the above approximation to compute $\nabla_{w_i} L_{reg}(w)$ for $i = 1, 2, \cdots, n$.

(b) (5 points) Find a piecewise solution to the following equation

$$\nabla_{w_i} L_{reg}(w) = 0,$$

for $i = 1, 2, \cdots, n$. Your solution should only be in terms of w_i^*, h_i, α.